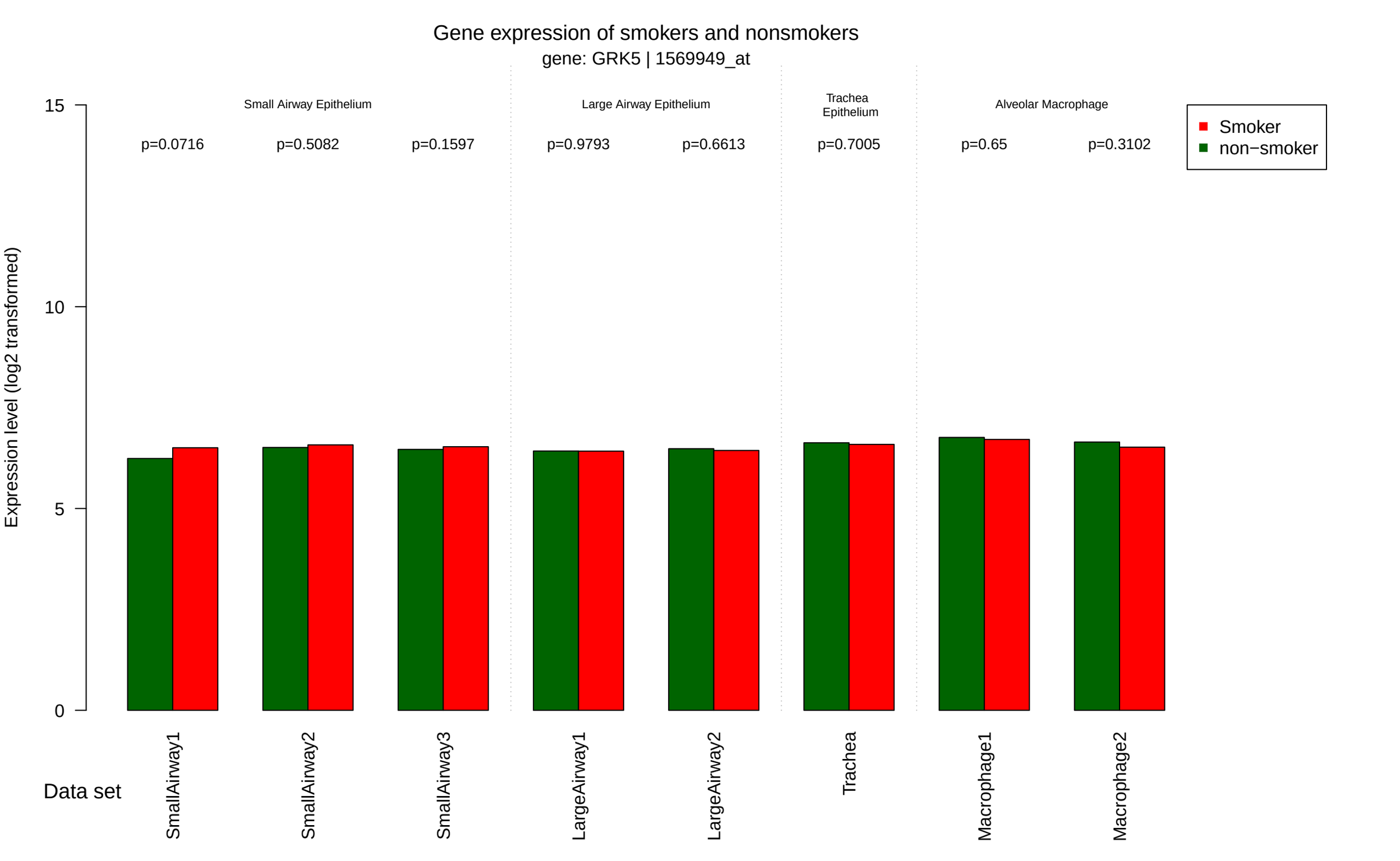
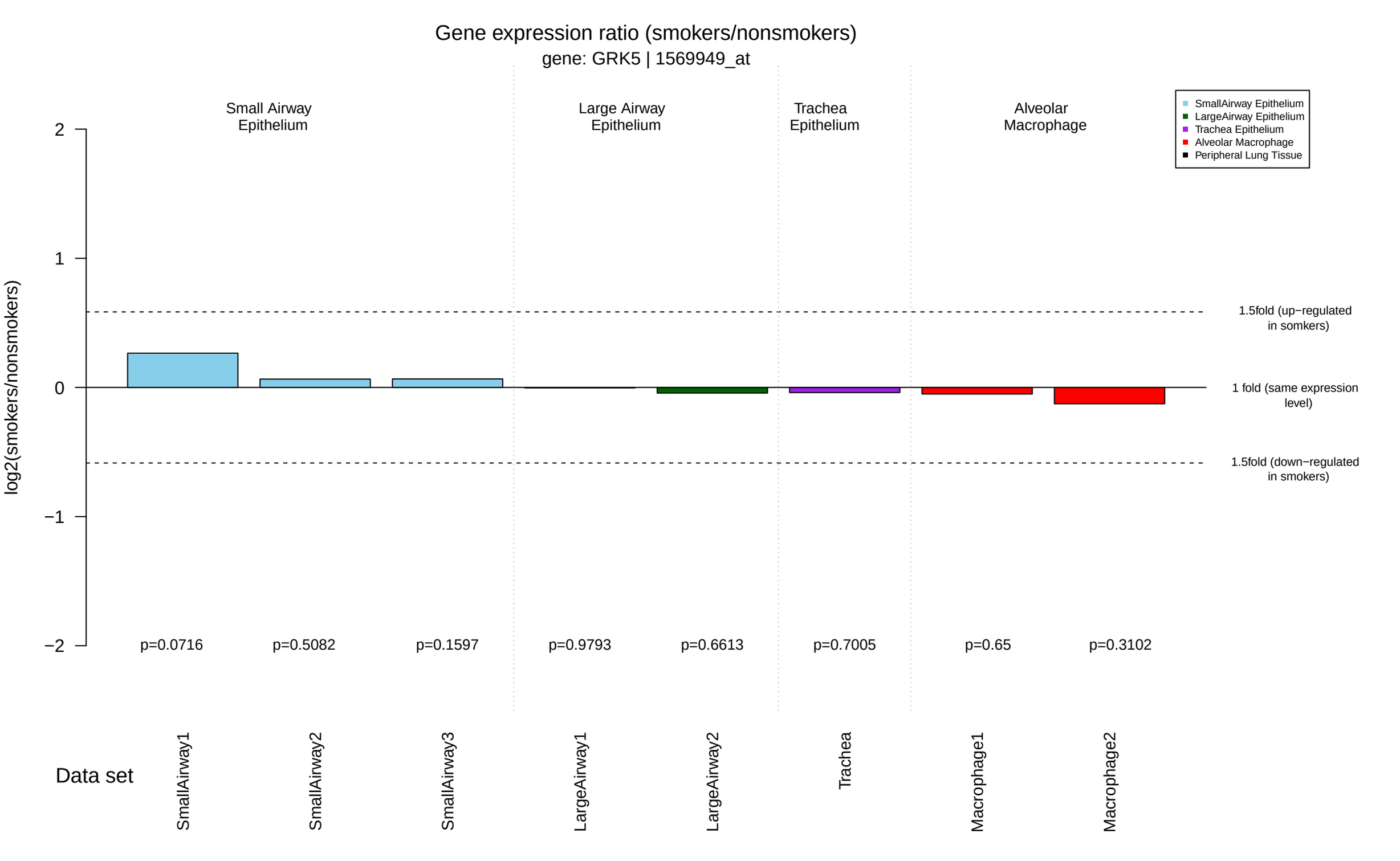
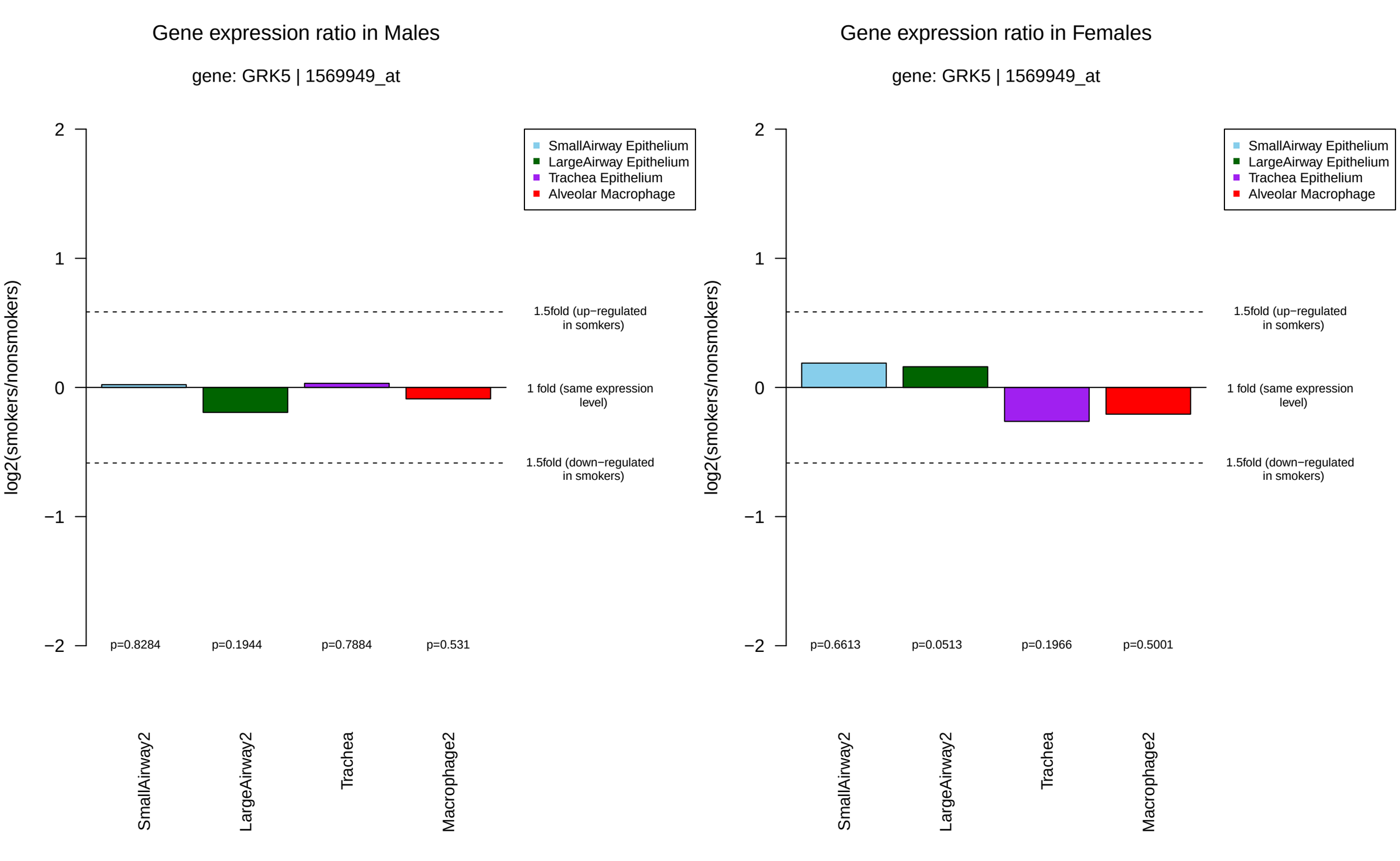
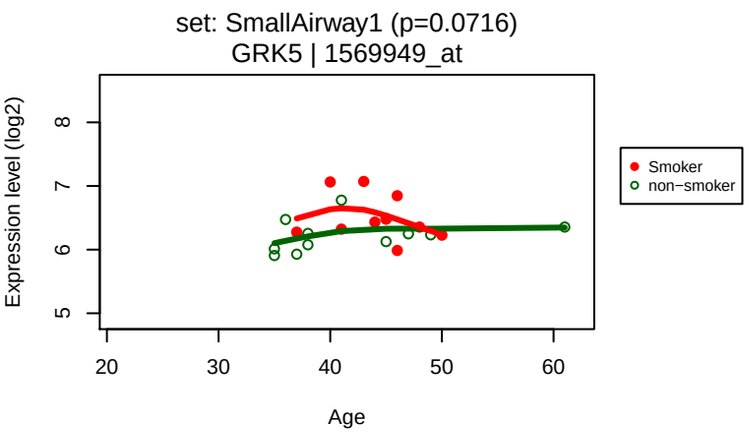
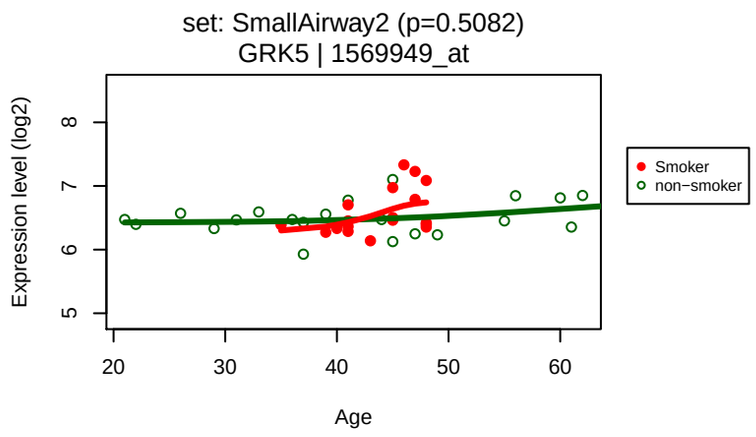
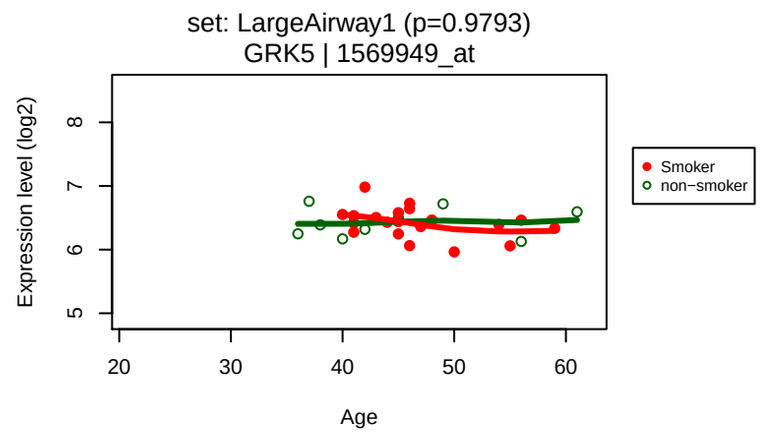

Gene expression of smokers and nonsmokers
gene: GRK5 | 1569949_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0716
p=0.5082
p=0.1597
p=0.9793
p=0.6613
p=0.7005
p=0.65
p=0.3102
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

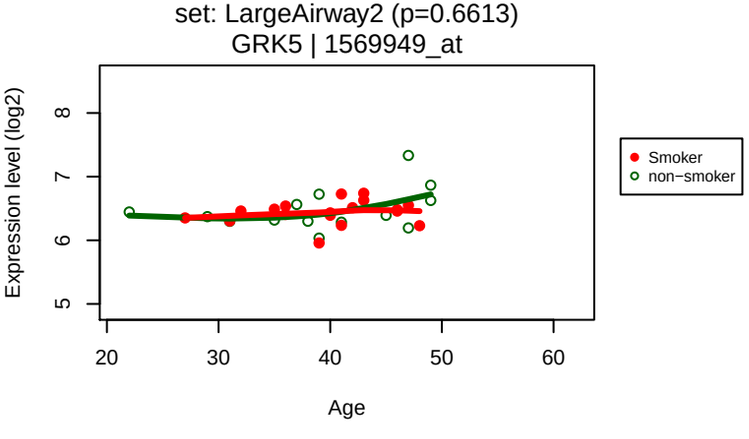

Gene expression ratio (smokers/nonsmokers)
gene: GRK5 | 1569949_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0716
p=0.5082
p=0.1597
p=0.9793
p=0.6613
p=0.7005
p=0.65
p=0.3102
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

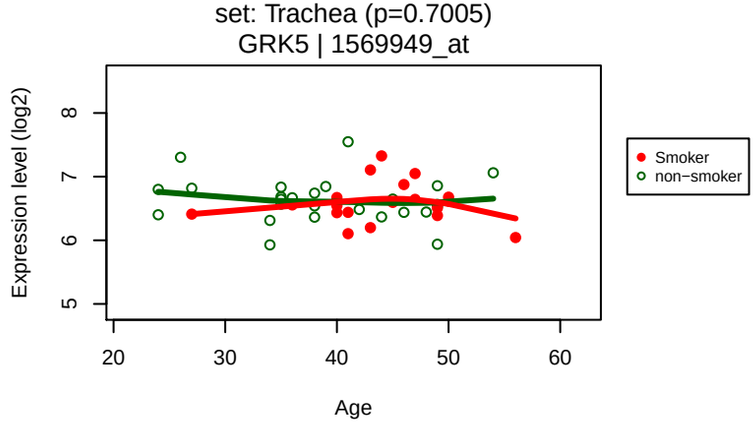

Gene expression ratio in Males
gene: GRK5 | 1569949_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8284
p=0.1944
p=0.7884
p=0.531
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRK5 | 1569949_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6613
p=0.0513
p=0.1966
p=0.5001
SmallAirway2
LargeAirway2
Trachea
Macrophage2

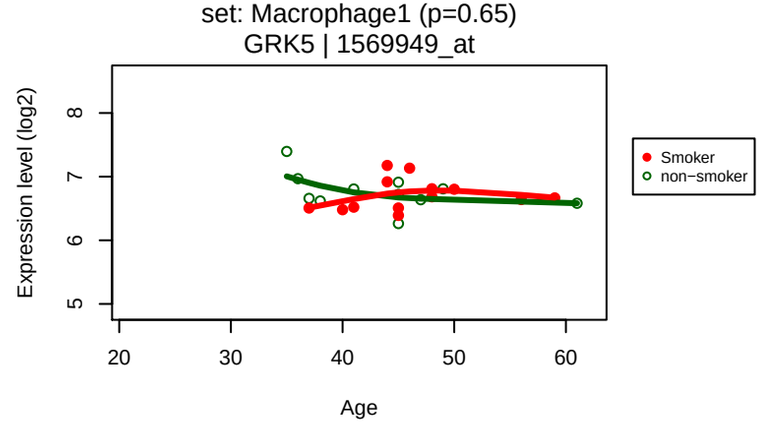

set: SmallAirway1 (p=0.0716)
GRK5 | 1569949_at
Expression level (log2)
Age
Smoker
non−smoker

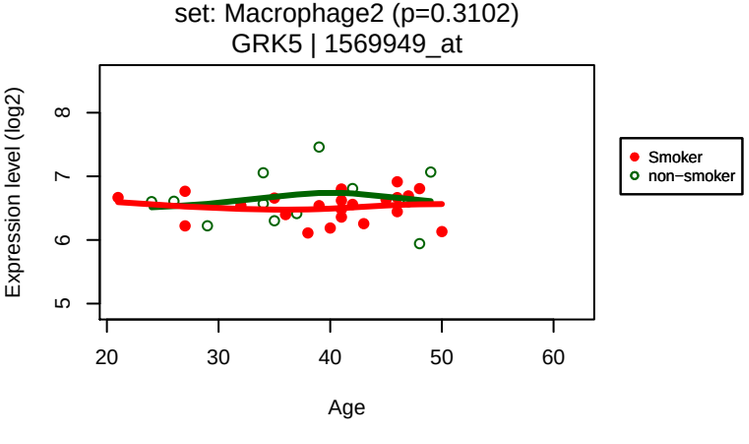

set: SmallAirway2 (p=0.5082)
GRK5 | 1569949_at
Expression level (log2)
Age
Smoker
non−smoker

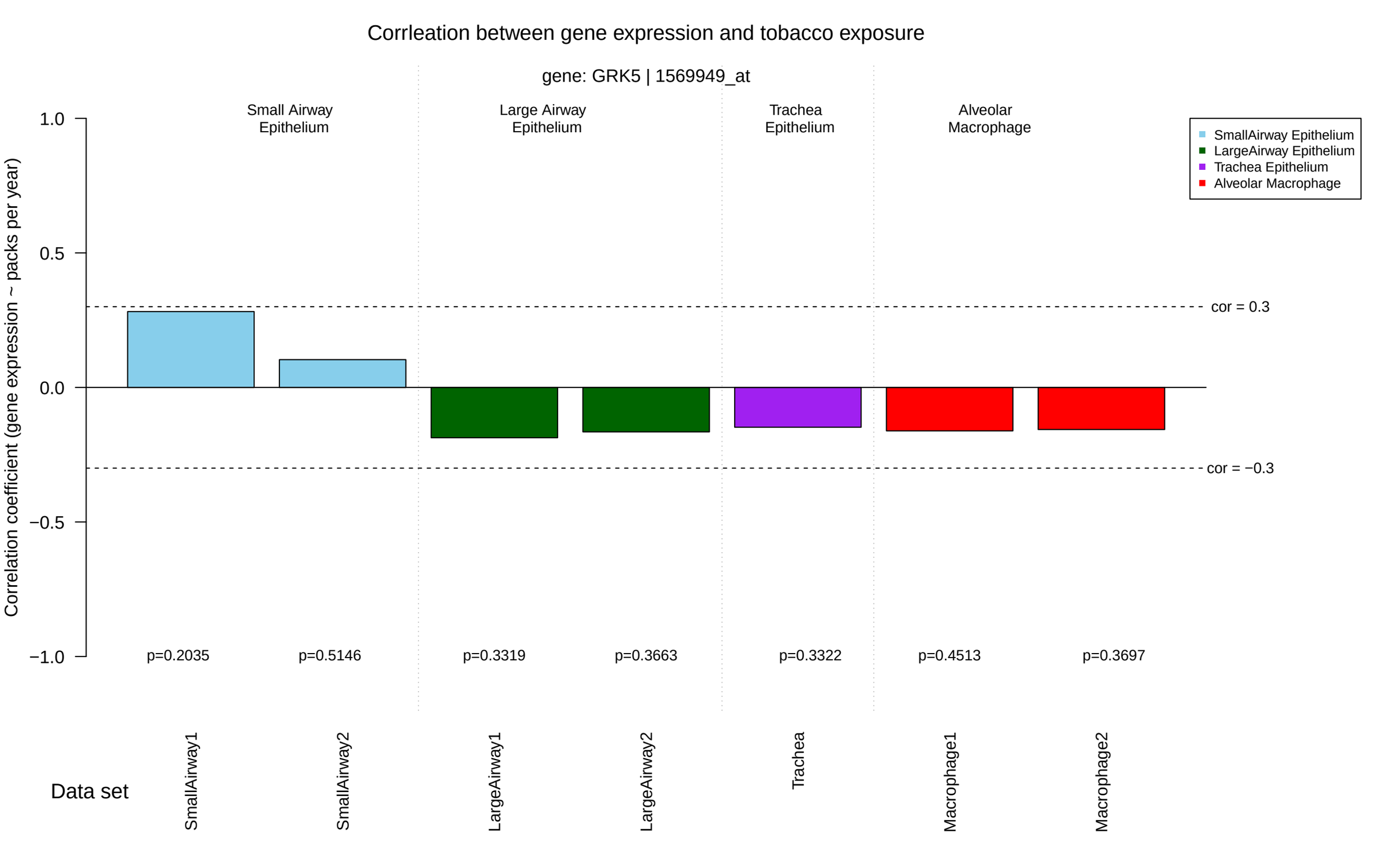

set: LargeAirway1 (p=0.9793)
GRK5 | 1569949_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6613)
GRK5 | 1569949_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7005)
GRK5 | 1569949_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.65)
GRK5 | 1569949_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3102)
GRK5 | 1569949_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRK5 | 1569949_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2035
p=0.5146
p=0.3319
p=0.3663
p=0.3322
p=0.4513
p=0.3697
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2